Practice Problems
Calculus Physics II

Chapter 45 Magnetism

1) The source of all magnetism is

   A) tiny pieces of iron.
   B) ferromagnetic materials.
   C) tiny domains of aligned atoms.
   D) moving electric charge.
   E) none of these

2) The units of the magnetic field are

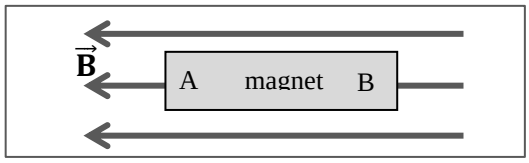
   A) Newtons
   B) Tesla
   C) Weber's
   D) Farad's

3) The conventional direction of magnetic field lines outside a magnet are from

   A) north to south.
   B) south to north.
   C) either way

4) A bar magnet is placed in a magnetic field and aligns in the field as shown, what are the poles of the magnetic indicated by the letters A and B?

   A) A is a north pole and B is a south pole.
   B) A is a north pole and B is a north pole.
   C) A is a south pole and B is a south pole.
   D) A is a south pole and B is a north pole.



5) Whereas electric charges can be isolated, magnetic poles

   A) can also be isolated.
   B) cannot be isolated.
   C) none of the above

6) If you break a bar magnet in half, you will

   A) destroy its magnetic properties.
   B) have two magnets.
   C) have two and a half magnets.
   D) none of the above

7) An iron rod becomes magnetic when

   A) its electrons stop moving and point in the same direction.
   B) positive ions accumulate at one end and negative ions at the other end.
   C) its atoms are aligned having plus charges on one side and negative charges on the other.
   D) the net spins of its electrons are in the same direction.
   E) none of these.

8) Which statement best answers the statement; Ferromagnetic material has a permeability that is

A) much greater than the permeability of air.
B) greater than the permeability of air.
C) the same as the permeability of air.
D) less than the permeability of air.
E) much less than the permeability of air.

9) Several paper clips dangle from the north pole of a magnet. The induced pole in the bottom of the lowermost paper clip is a

A) north or south pole – no difference really.
B) south pole.
C) north pole.

10) When a ferromagnetic material is placed in an external magnetic field, the net magnetic field of its magnetic domains becomes

A) smaller.
B) zero.
C) larger.

11) When a compass aligns in Earth's magnetic field, the compose points to

A) the magnetic north pole.
B) the magnetic south pole.
C) either the magnetic north or the magnetic south pole depending on the compose.

12) About what is the magnetic field strength at the surface of the earth?

A) 3×10^{-5} T
B) 3×10^{5} T
C) 3×10^{2} T
D) 3×10^{9} T
E) 3×10^{-2} T

KEY

1) D
2) B
3) A
4) A
5) B
6) B
7) D
8) A
9) C
10) C
11) B
12) A